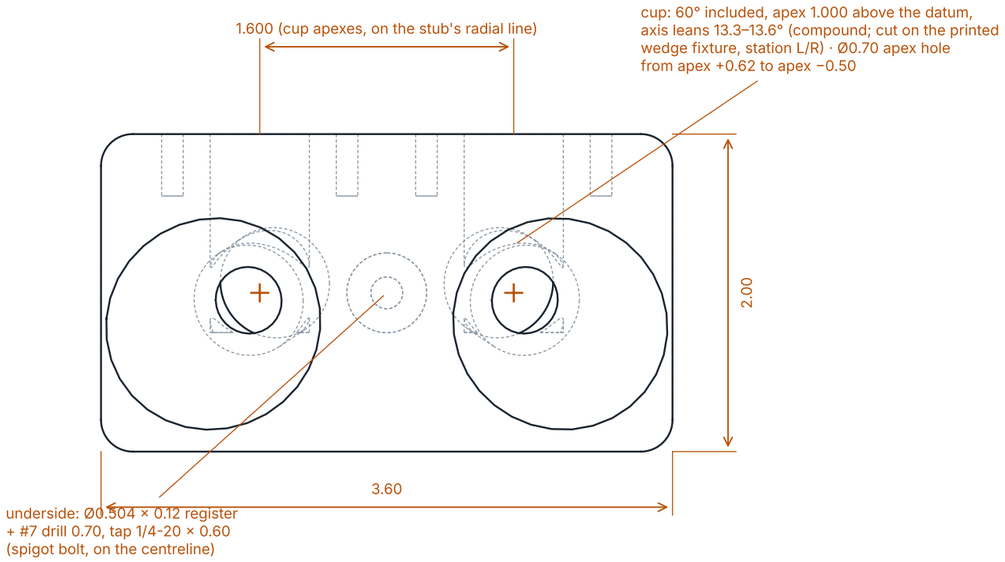
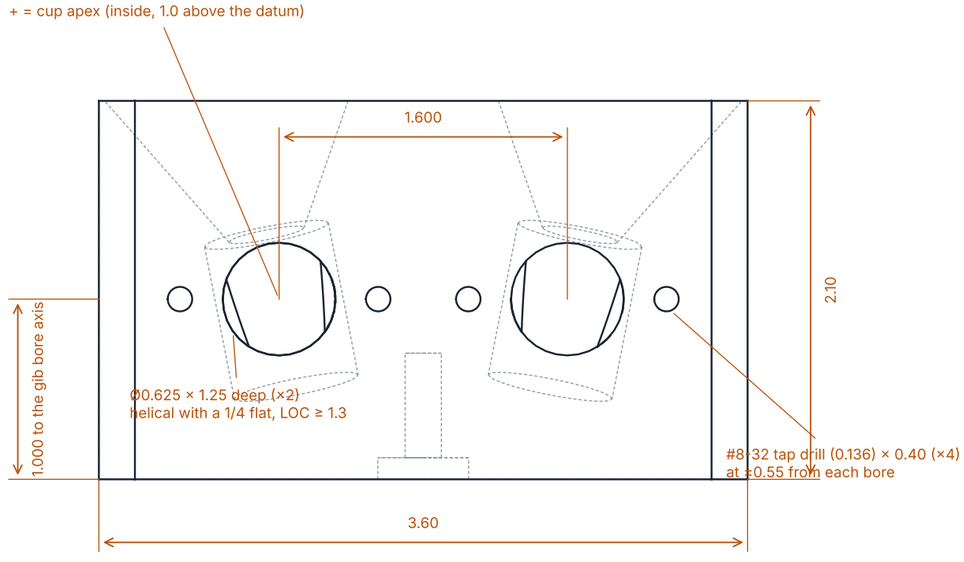
16" GoTo Dob rebuild — Socket block (truss cone sockets, gib clamps)

6061-T6 from 2-1/4 sq bar, faced 3.60 t × 2.00 r × 2.10 u · qty 3 (point + 2 back; 210 = 330) · inches · from the CadQuery model, 2026-11-08 · sheet 1/4

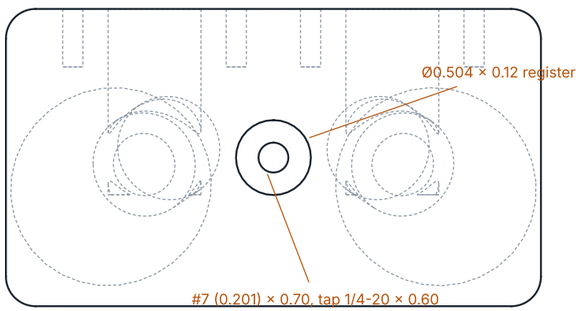
TOP (cup face). Outside face at the top of the view; x = tangential



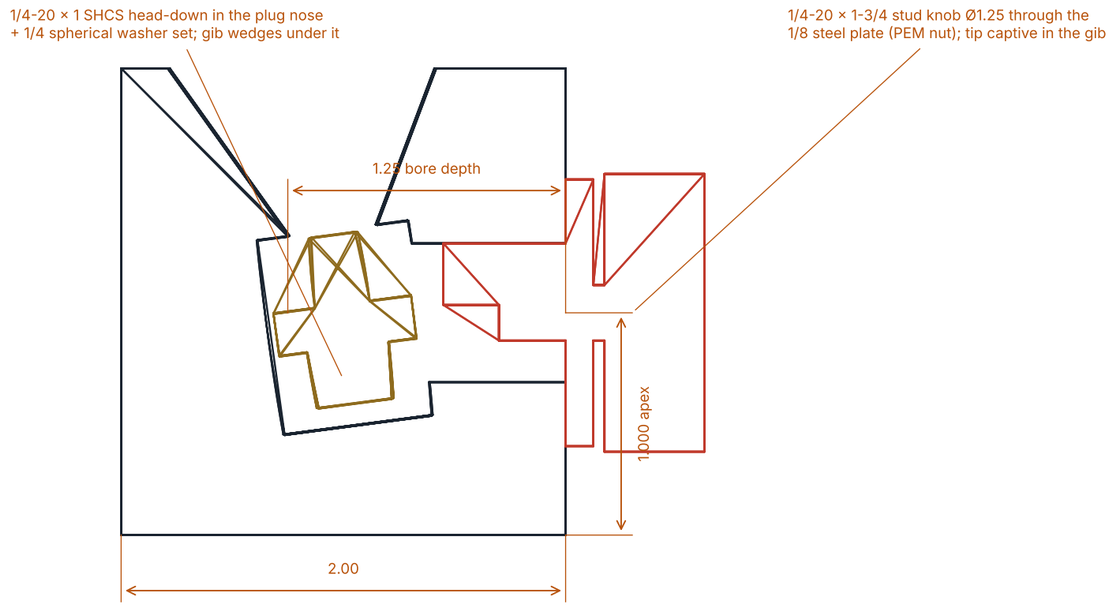
OUTSIDE FACE (Op2, block on its inside face, no fixture)



UNDERSIDE (Op5): register + spigot bolt tap



SECTION through the left cup, radial plane (x = tangential −0.8): +z = outside face. red = gib, plate, knob; brown = head screw

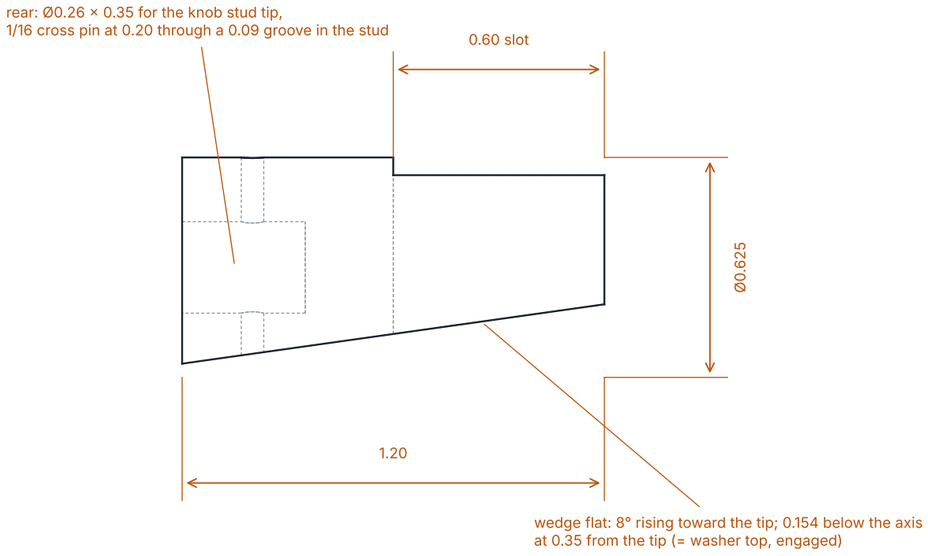


Datum: stub top centre (block underside centre). t = tangential, r = radial (+ = outside face), u = up.
Op1 square the stock · Op2 outside face up: gib bores + tap drills (socket_op2_gib_bores.nc) · Op3/Op4 cups + apex holes on the printed wedge fixtures (socket_*_L/R.nc) · Op5 underside: register + tap · tap #8-32 ×4 and 1/4-20 ×1 by hand.
Cone contact after the apex hole: r 0.35–0.62 on the 60° cup. The screw leans 13° while the gib is horizontal: spherical washer, not flat.

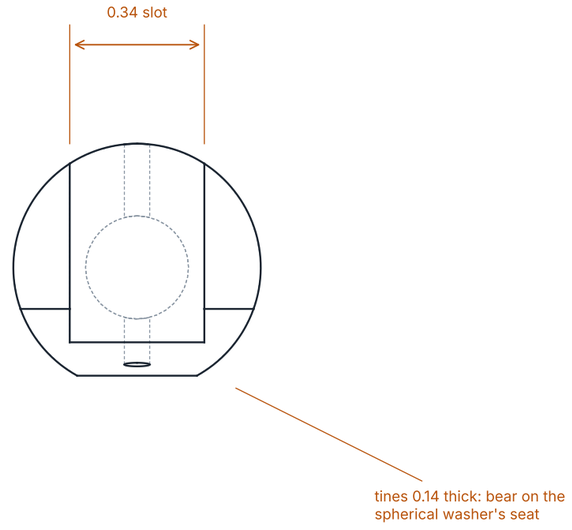
16" GoTo Dob rebuild — Gib pin and retainer plate (gib clamp)

gib: O1 drill rod 5/8; plate: 1/8 steel · qty 6 + 6 · inches · from the CadQuery model, 2026-11-08 · sheet 2/4

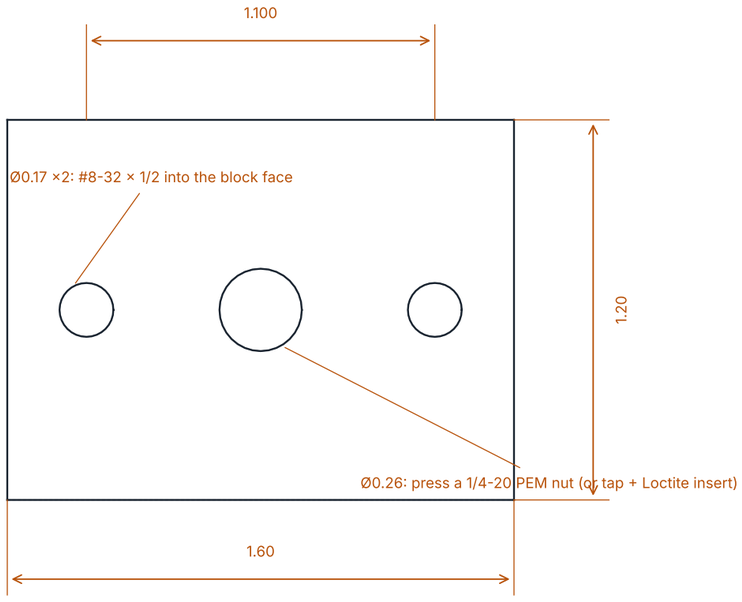
GIB, side view (tip at the right; wedge flat on the underside; slot through, vertical)



GIB, end view from the tip



RETAINER PLATE, face view

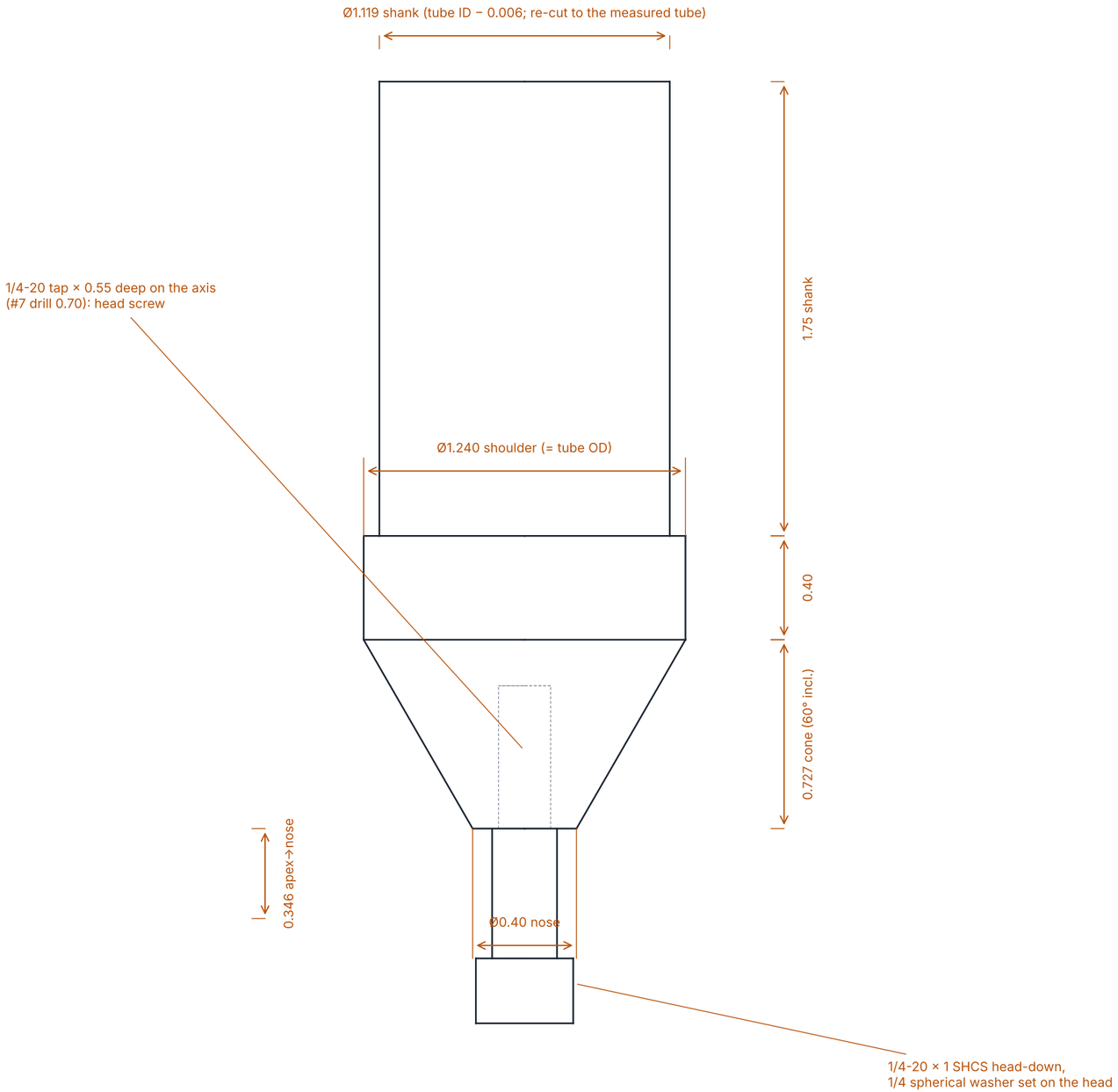


Gib: cut 1.20 from 5/8 drill rod; mill the 0.34 slot 0.60 deep from the tip and the 8° flat along the whole underside with the rod in a V-block on the AltMill (flat 0.105 below the axis at the tip, 0.154 at 0.35 from the tip = the contact when engaged, 0.27 at the rear). The round top and sides guide it in the bore. Released: tip 0.80 in from the face (head passes). Engaged: tip ~1.15 in. Travel 0.35 = ~7 turns of the 1/4-20 knob. If a tightened joint will not release with a tap after the knob is backed off, recut the wedge at 10° (20% less pull, still > 150 lb).

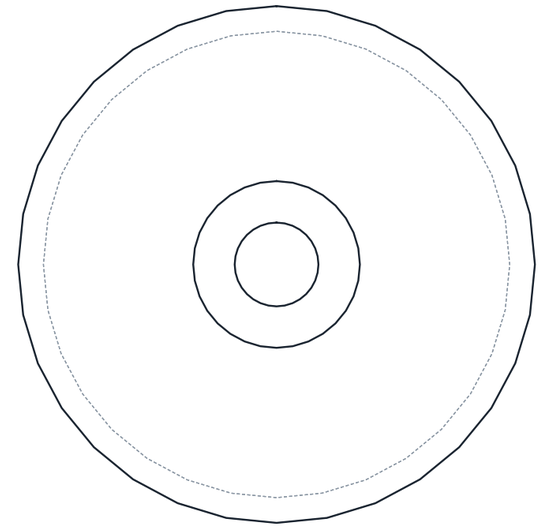
16" GoTo Dob rebuild — Cone plug (bonded into each carbon pole's base end)

6061 from 1-3/8 round bar (Vortex rotary, continuous spiral) · qty 6 (+ 6 pair-end plugs are plain spigots) · inches · from the CadQuery model, 2026-11-08 · sheet 3/4

SIDE VIEW, pole axis vertical; cup apex at y = 0; head screw fitted



END VIEW from the nose

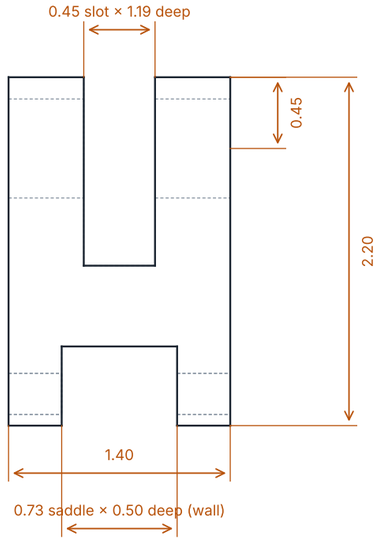


Turn as one continuous spiral on the Vortex (A one way, X/Z following nose → cone → shoulder → shank); rough in depth passes, one finish pass.
Bond 1.75 shank into the tube with structural epoxy (DP420); the shoulder stops against the tube end. Cone seats in the 60° cup from r 0.35 to 0.62.
The geometric apex (y = 0) is the truss node; the nose face sits 0.346 above it. Pole node-to-node 37.30 / 37.35 sets the tube cut length (35.87 / 35.92).

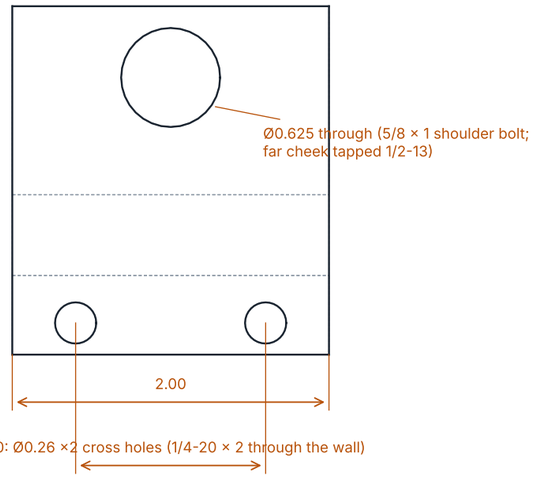
16" GoTo Dob rebuild — Roller blocks (alt: saddle on the wall; az: hung through the floor)

6061 from 1-1/2 × 2 bar, faced 1.40 wide · qty 4 alt + 4 az · inches · from the CadQuery model, 2026-11-08 · sheet 4/4

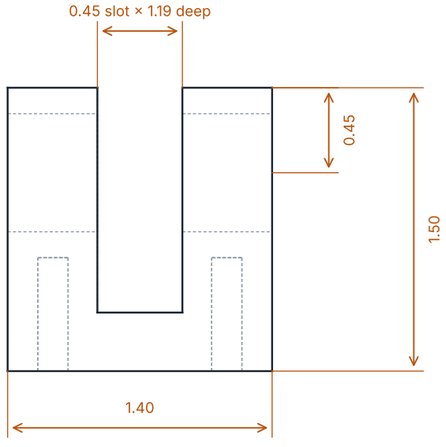
ALT block: end view (X = roller axis)



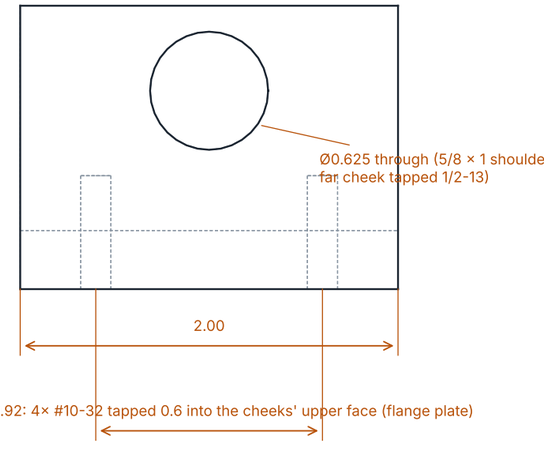
ALT block: side view (Y = along the track)



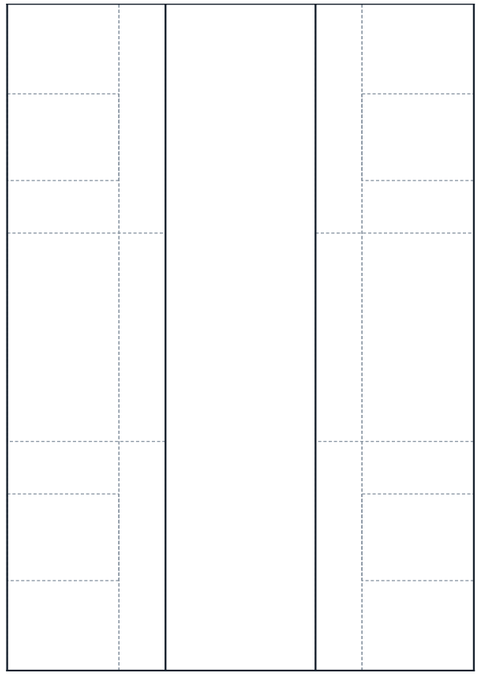
AZ block: end view (X = roller axis)



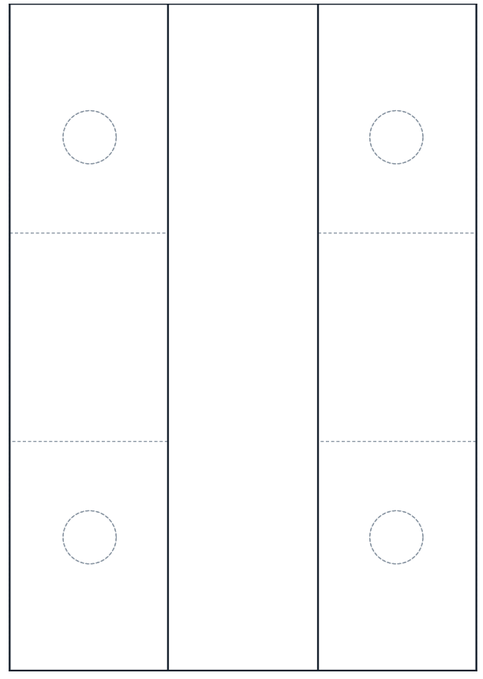
AZ block: side view (Y = along the track)



ALT block: top (slot and roller)



AZ block: top (slot and roller)



Frame: X = roller axis (block width 1.40), Y = tangential along the track (2.00), Z = toward the track. Roller R10 5/8 × 1-3/8 × 11/32 on a 5/8 × 1 shoulder bolt, bolt head outboard (alt) / radially outboard (az, the floor cutout is 0.5 wider on that side). Roller protrudes 0.24 past the block top.
Alt: block saddles the wall top in its notch (wall top at R + 2.0), two 1/4-20 × 2 cross bolts; 1/8 Al + 1/16 PTFE tab on the outer face as the lateral stop.
Az: roller side DOWN; block hangs through a 1.94 × 2.04 floor cutout, 0.79 below the floor, under a 1/8 flange plate (2.4 × 3.0) on 4× #10-32.